

a)

Nech $B = \{\vec{u}_1, \vec{u}_2, \vec{u}_3\}$ je výsledok G-S ortogonalizácie postupnosti $\{1, x, x^2\}$

$\vec{u}_1$ dostaneme vyhormovaním 1.

t.j. $\vec{u}_1 = \frac{1}{\|\vec{1}\|}$, kde $\|\vec{1}\| = \sqrt{\int_1^5 1 \cdot 1 dx} = 2$

$$\vec{u}_1 = \frac{1}{2}$$

ďalej $\vec{u}_2 = \frac{x - \vec{w}_1}{\|x - \vec{w}_1\|}$, kde $\vec{w}_1$ je OG projekcia x na $\frac{1}{2}$

t.j. $\vec{w}_1 = \langle \frac{1}{2}, x \rangle \frac{1}{2} = \frac{1}{2} \int_1^5 x \cdot \frac{1}{2} dx = 3$

$$\|x - 3\| = \sqrt{\int_1^5 (x-3)^2 dx} = \frac{4}{\sqrt{3}}$$

$$\vec{u}_2 = \frac{\sqrt{3}}{4} (x-3)$$

podobne nájdeme $\vec{u}_3 = \frac{x^2 - \vec{w}_2 - \vec{w}_3}{\|x^2 - \vec{w}_2 - \vec{w}_3\|}$

$$\vec{w}_2 = \langle \frac{\sqrt{3}}{4} (x-3), x^2 \rangle \cdot \frac{\sqrt{3}}{4} (x-3) = \frac{\sqrt{3}}{4} (x-3) \int_1^5 \frac{\sqrt{3}}{4} (x-3) x^2 dx = 6(x-3)$$

$$\vec{w}_3 = \langle \frac{1}{2}, x^2 \rangle \cdot \frac{1}{2} = \frac{1}{2} \int_1^5 \frac{1}{2} x^2 dx = \frac{31}{3}$$

$$\|x^2 - 6(x-3) - \frac{31}{3}\| = \sqrt{\int_1^5 (x^2 - 6x + \frac{23}{3})^2 dx} = \frac{16}{3\sqrt{5}}$$

t.j. $\vec{u}_3 = \frac{16}{3\sqrt{5}} (x^2 - 6x + \frac{23}{3})$

b)

Označme OG projekciu $\vec{v} = ax^2 + bx + c = a\vec{u}_3 + b\vec{u}_2 + c\vec{u}_1$

keďže B je ON a, b, c nájdeme ako sk.s. prvkov B so $\sin x$.

$$a' = \langle \vec{u}_3, \sin x \rangle = \int_1^5 \frac{16}{3\sqrt{5}} (x^2 - 6x + \frac{23}{3}) \sin x dx = -0,71$$

$$b' = \langle \vec{u}_2, \sin x \rangle = \int_1^5 \frac{\sqrt{3}}{4} (x-3) \sin x dx = -1,49$$

$$c' = \langle \vec{u}_1, \sin x \rangle = \int_1^5 \frac{1}{2} \sin x = 0,13$$

teda $\vec{v} = -0,71 \cdot \frac{16}{3\sqrt{5}} (x^2 - 6x + \frac{23}{3}) - 1,49 \cdot \frac{\sqrt{3}}{4} (x-3) + 0,13 \cdot \frac{1}{2}$

$$\vec{v} = -1,69x^2 + 9,51x - 10,98$$

$$\Rightarrow a = -1,69$$

$$b = 9,51$$

$$c = -10,98$$

c)

$$\|\sin x - \vec{v}\| = \sqrt{\int_1^5 (\sin x + 1,69x^2 - 9,51x + 10,98)^2 dx} = 6,95$$